Project Documentation: Packaged Commodity Compliance & Food Safety Platform

(Smart India Hackathon 2026 — Team Draft)

📌 **FINALIZED TRACK & SCOPE (Read This First)**

- **We are registering under Student Innovation — SIH26197 (AICTE, theme: Agriculture, FoodTech & Rural Development).**
- This is a **self-proposed idea track** — there is no fixed government rubric we must satisfy, unlike an official Problem Statement.
- We found that our Legal Metrology concept also matches an **official PS: SIH26034** (Ministry of Consumer Affairs, Food & Public Distribution — "Software System to check compliance of Packaged Commodities under Legal Metrology Rules, 2011 by scanning products, images and labels"). We are **not** registering under this PS — we are only borrowing its problem framing as inspiration for our Legal Metrology module.
- Because we're in Student Innovation, **we have full freedom to shape the app our way** — Food Safety is our main, lead feature, and we are **keeping the Officer section by choice (not obligation)** because it adds an impressive, technically strong touch to the project — not because any rubric requires it.

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1. What Are We Building? (Simple Explanation)

We are building a **web app that scans product labels** (like the ones on packaged food, cosmetics, or any packaged item) and tells you two things:

1. **Is this product legally compliant?** — Does it have the correct MRP, quantity, manufacturing date, and other details required by law? *(Legal Metrology check)*
2. **Is this product safe/healthy for me?** — What are the ingredients, are any of them harmful, is it expired? *(Food Safety check)*

If something is wrong, the user can **file a complaint** with proof (photos, ID verification), and it gets sent to the correct government department automatically.

In one line: *Scan a product label once → know if it's legally compliant AND safe to eat → report violations directly to the right authority, with proof, so false complaints are minimized.*

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2. Why This Idea? (Background)

- We started by exploring SIH themes and eventually picked **FoodTech**.
- Our first idea was a **food complaint app** — but we discovered FSSAI already has an app called **"Food Safety Connect"** that does complaint filing.
- Research showed FSSAI's app has a **real, documented problem**: even though the app exists, many FDA officials never received any complaints through it because there was no operational process connecting the app to actual enforcement action. The app is also reported to be slow and hard to use.
- This meant: don't compete with FSSAI's complaint feature — instead, add something **they don't have**: an **ingredient/health scanner** with **verified, evidence-backed complaints**.
- Separately, we found an **official SIH Problem Statement (PS)** about **Legal Metrology compliance** (MRP, net quantity, manufacturing date, labelling rules) — a completely different but related law that applies to all packaged goods (not just food).
- Since both Legal Metrology and Food Safety rules apply to the **same product label**, we explored combining both into **one scanning engine**.

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3. The Two Laws Involved (Keep This Simple)

Law	What it Checks	Who Enforces It
Legal Metrology (Packaged Commodities) Rules, 2011	MRP, net quantity, manufacturing date, manufacturer address, font size on label	Department of Consumer Affairs
FSS (Labelling & Display) Regulations, 2020	Ingredients, allergens, FSSAI license number, nutrition info	FSSAI (Food Safety Authority)

Both are checked from the **same photo of a product label**, which is why one shared scanning engine makes sense technically — even though the two laws belong to **different government departments**.

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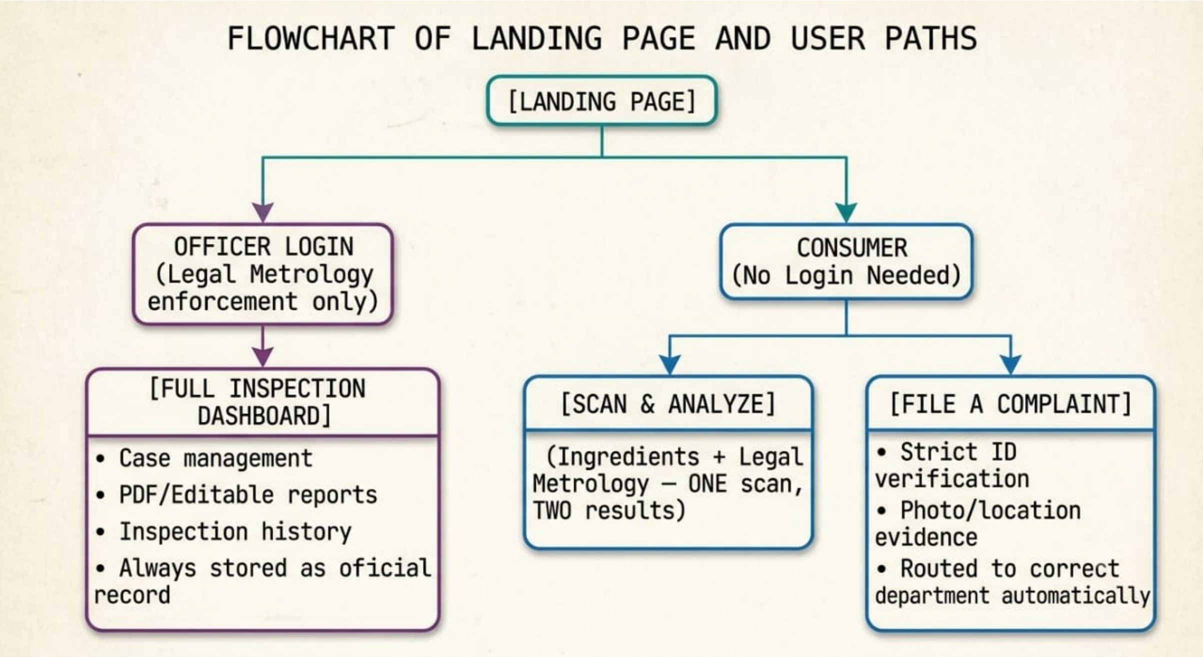
4. Final App Structure (As Decided So Far)



✅ **RESOLVED:** We initially debated whether Officer and Consumer sections should even be in the same app, since they serve completely different users. We've now confirmed (see Section 11 below) that **the consumer's compliance check does NOT technically depend on officer data at all** — it works standalone using the 2011 Rules as a rule engine. The officer section is being **kept anyway, by choice**, scoped small, purely because it makes the project look more complete and impressive — not because it's structurally required. Full reasoning is in Section 11.

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5. Core Features Explained Simply

A. One Scan, Two Results (Consumer Side)

- User uploads **one photo** of the product label.
- OCR (text-reading AI) extracts all the text.
- The system automatically identifies which part is the **MRP/quantity/date section** and which part is the **ingredients section** — no manual circling needed by the user.
- Two results are shown together:
  - ✅/❌ Legal Metrology compliance (MRP, quantity, date, address present or not)

    - ⚠️ Food Safety analysis (harmful ingredients, allergens, health score, expiry check)

B. Ingredient Scanner (Food Safety)

- Extracts ingredient list from the label.
- Compares each ingredient against a database of harmful/banned/restricted substances.
- Gives a simple **health score** (like Nutri-Score) with plain-language explanations (e.g., "Contains Tartrazine — linked to hyperactivity in children").
- Can be **personalized** — e.g., extra warnings if the user has allergies or diabetes.

C. Officer Section (Kept by Choice — Not Required by Any Rubric)

- For enforcement officers only (login required).
- **Scoped small and sharp**, not a full enforcement suite: a simple login + "Log an Inspection" form (shop name/location, product scanned, compliant/violation, date).
- Every inspection scan (pass or fail) is **always recorded** — this becomes a compliance record in our own database.
- A citizen-facing **"Shop Compliance History" lookup** is added on the consumer side — search a shop, see past inspection results + resolved complaints. This is the one genuine (optional) connection between the two sections.
- We are deliberately skipping heavier features (PDF/editable report exports, case escalation workflows, officer role hierarchies) unless time allows, since they don't add to the "impressive touch" and cost significant build time.

D. Complaint Filing System

- Consumers can file a complaint if they find a violation (expired, mislabeled, missing MRP, etc.).
- Complaints require **evidence**: photos, shop location, product details, and **identity verification** (explained below) to prevent false/fake complaints.
- **Complaints are automatically routed to the correct government department** based on which law was violated:
    - Legal Metrology issue → Department of Consumer Affairs
    - Food Safety issue → FSSAI
    - If both are violated, the complaint is **split and sent to both departments**.

- Complaints are **prioritized** based on severity, evidence strength, how long they've been pending, and whether the same shop has multiple complaints.

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6. Identity Verification for Complaints (To Prevent False Complaints)

We are offering **multiple verification methods** so no one is blocked from filing a complaint just because they don't have a specific document:

Method	Trust Level	Notes
Aadhaar QR Code Scan	High	Scan the QR on the Aadhaar card/e-Aadhaar — verifies identity WITHOUT storing the Aadhaar number (this is legally required, explained below)
DigiLocker Login	High	User logs into their own DigiLocker account and shares verified documents directly — consent is built into the login process itself
PAN Card (OCR)	Medium	Extract and validate PAN number format
Voter ID / Driving License (OCR)	Medium	Extract ID number and details
Phone OTP only	Low (fallback)	Used only if no document is available — lowest trust tier

⚠️ Important Legal Note (Why We Can't Store Aadhaar Numbers)

- **The Aadhaar Act legally prohibits storing the Aadhaar number itself**, even if the user gives consent — consent does NOT override this law.
- The correct, government-approved method is to **scan the QR code** on the Aadhaar card, verify its digital signature (proves it's not fake/tampered), extract only name/address/photo, and **never store the Aadhaar number** — only a "verified: yes/no" status.
- For PAN, Voter ID, and Driving License, we must follow the **Digital Personal Data Protection Act (DPDP) 2023**, which requires:
    - Clear, specific consent before collecting any ID data
    - Only using the data for the stated purpose (verifying the complaint)
    - Allowing users to request deletion of their data

- Masking sensitive numbers when displaying them anywhere in the app

This shows judges we understand real data protection law — something most student projects overlook.

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7. Database Logic (Where Data Goes)

- **Officer scans** → Always saved as an official Legal Metrology inspection record (whether compliant or not) — sent to the Legal Metrology / Consumer Affairs system.
- **Consumer's personal scans** (just checking a product, not complaining) → Stays private, never sent anywhere.
- **Consumer complaints** → Only created when the user actively files one. Gets verified, then routed to the correct department(s) based on the type of violation found.
- For the hackathon demo, since we don't have real access to government systems, we will **simulate** this routing (build mock dashboards for each department) and explain that in production, this would connect to real government APIs.

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8. Build Priority Plan (Time Management for Hackathon)

1. **Shared core first** — OCR + text extraction engine (works for both Legal Metrology and Food Safety)
2. **Food Safety module** — ingredient scanner, health scoring, expiry check (our team's main interest)
3. **Legal Metrology module** — MRP/quantity/date checks (needed if we're required to follow the official PS)
4. **Complaint system** — verification, evidence upload, routing logic, priority scoring
5. **Dashboards** — consumer results screen + (if needed) officer dashboard
6. **Polish + demo script**

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9. Key Questions & Answers (Important Decisions So Far)

Q: Why can't we just build a normal food complaint app? A: FSSAI already has one (Food Safety Connect). We need to differentiate — our ingredient scanner and evidence-verified complaints are the unique value, not the complaint filing itself.

Q: Why would people bother scanning every product? A: They won't scan everything — realistically, people scan when in doubt (buying for a child, checking for allergies, unfamiliar brand, suspicious product). We should not claim universal scanning habits in our pitch.

Q: If a product is expired, why not just tell the shopkeeper instead of filing a formal complaint? A: Telling the shopkeeper only fixes that one moment — no record, no accountability, and the next customer faces the same risk. A formal, evidence-backed complaint creates a paper trail that can lead to real regulatory action and protects future customers too.

Q: Can we combine Legal Metrology and Food Safety into one project? A: Yes — it's valid and can be a genuine strength if framed correctly: one shared scanning engine, because a single product label legally has to comply with multiple regulations at once. But our official grading (if using the Legal Metrology PS) is based on that PS specifically — Food Safety should be positioned as a bonus/extension, not an equal second project, UNLESS we confirm we can self-propose our own idea instead.

Q: Can we store Aadhaar card details since the user gives consent? A: **No.** The Aadhaar Act legally prohibits storing the Aadhaar number even with consent. We must use the official QR-code-based offline verification method instead, and never store the Aadhaar number itself.

Q: Should the officer (Legal Metrology) section and consumer (Food Safety) section really be one app? A: This is still an open decision. They serve genuinely different users with no direct overlap. We are deciding between: (A) keep both, with Food Safety as the lead feature and a smaller officer module for PS eligibility, or (B) drop the officer section completely and make Food Safety the sole focus — but this depends on whether our SIH round requires an official Problem Statement or allows self-proposed ideas. **This needs to be confirmed with our SPOC/faculty coordinator before we finalize scope.**

Q: Where does officer data go vs. consumer complaint data? A: Officer inspection data always goes to our own database as a compliance record. Consumer complaints are routed based on the type of violation — Legal Metrology issues to Consumer Affairs, Food Safety issues to FSSAI, and if a complaint has both types of violations, it gets split and sent to both departments (simulated for the hackathon demo, since we don't have real access to government systems — see Section 11).

Q: Can't we use the actual government's existing Legal Metrology data instead of building our own database? A: No — this data doesn't exist in a usable form yet. The government is building a unified portal called **eMaap**, but it's still under development, and enforcement/inspection records are not digitized or centralized anywhere today — even state governments each run separate, disconnected portals. There is nothing to "fetch" — we have to generate and own this data ourselves. (Full details in Section 11.)

Q: If we use the 2011 Rules to check products, why do we need officer-collected data at all? A: We don't — for the core compliance check. The consumer's scan works completely standalone: it reads the label and checks it against the 2011 Rules (the legal rulebook, not a database) in real time. Officer data is only useful for one optional bonus feature: showing a shop's past inspection/violation history. It is not required for the scanning feature to function. (Full details in Section 11.)

Q: If a consumer visits a shop that no officer has ever inspected, will they still get information? A: Yes — because the compliance check is a live, independent analysis of that specific label, not a lookup of past inspection records. It works the same whether or not that shop has ever been officially visited.

Q: Should we still keep the officer section, given it's not technically required? A: Yes, by team decision — not because it's required, but because it demonstrates real technical depth (auth, structured data collection, cross-linking with citizen lookups) and mirrors a real government gap (eMaap doesn't exist yet, so we're showcasing a working version of what the government itself is still building). Kept scoped small so it doesn't eat into Food Safety build time.

Q: What does a real Legal Metrology officer actually do? A: Two separate jobs — only one of which is relevant to us:

1. **Instrument verification/stamping** (weighing scales, petrol pumps, taxi meters) — physically testing equipment for accuracy. *Not relevant to our app.*
2. **Packaged commodities inspection** — checking net quantity/labelling at manufacturer/packer premises and retail shops, through routine and surprise inspections, then compounding offences (fine) or sending serious cases to court. *This is the function our app supports.*

Q: Do officers have to check every single shop in India? A: No — real enforcement is sampling-based, not exhaustive. Officers cover a specific jurisdiction/zone and conduct periodic market inspections, surprise checks, and complaint-driven visits — never a full nationwide sweep. Our app doesn't need to solve "check every shop" either; it just needs to help officers prioritize which shops to check next (e.g., ones with pending citizen complaints).

Q: Does our Legal Metrology idea match an official SIH Problem Statement? A: Yes — **SIH26034** (Ministry of Consumer Affairs, Food & Public Distribution) matches our Legal Metrology concept almost exactly. However, we are **not** registering under this PS.

We're registered under **Student Innovation (SIH26197)**, which is self-proposed and theme-based, not PS-based — so we are not bound by SIH26034's specific rubric, even though we're inspired by it.

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11. Deep-Dive: Officer Data vs. Consumer Data (Why We Kept Both Sections)

This section documents the detailed reasoning that led to our final architecture decision, in case anyone on the team wants the full logic later.

11.1 The government's own Legal Metrology data isn't ready yet

We researched whether we could just pull real inspection/verification data from the government instead of building our own. We found:

- The Department of Consumer Affairs is building a new unified portal called **eMaap**, meant to integrate all State Legal Metrology Departments into one national system — **but it's still under development.**
- Right now, **each state runs its own separate portal**, and **enforcement activities are not digitized/online anywhere yet**, even within the government's own systems.
- **Conclusion:** there is no existing API or dataset to fetch shop-verification data from. We must generate this data ourselves through our own officer section. This is actually a strong pitch point — we're demonstrating a working version of something the government hasn't finished building yet.

11.2 The consumer scan does NOT depend on officer data

This was an important realization during our discussion:

- The consumer-facing compliance check works by reading a label and checking it against the **2011 Rules (the published legal requirements)** — this is static legal text, not a database of past records.
- This means the check is **live and independent** — it produces an accurate result for any product, at any shop, whether or not that shop was ever inspected by an officer.
- **Conclusion:** officer data is not required for the core scanning feature to work. It's optional, bonus information — specifically, a "shop compliance history" lookup — not core plumbing.

11.3 So why keep the officer section at all?

Because it's still valuable, even though it's optional:

- It fills a real, documented gap (no digitized enforcement records exist anywhere yet in India).
- It adds a genuine "shop history" trust feature for consumers (e.g., "This shop has had 2 confirmed violations in the last 6 months").
- It shows stronger technical range in our submission (authentication, structured government-style data entry, cross-linking between sections) — which helps at Student Innovation judging, where technical depth is a scoring factor even without a fixed rubric.
- **Decision:** Keep it, but scoped small — a basic officer login + inspection logging form, and a simple consumer-side shop lookup. Skip heavier enforcement features (case management, PDF exports, escalation workflows) to protect time for our main focus: Food Safety.

11.4 What a Legal Metrology officer actually does (for accuracy in our officer module)

- **Verification/stamping of instruments** (weighing scales, fuel pumps, taxi meters) — not part of our app.
- **Packaged commodities inspection** — checking net quantity, labelling, and declarations at shops/manufacturer premises via routine and surprise inspections.
- If a violation is found: officer either **compounds the offence** (fine/settlement) or, if serious/unresolved, **sends the case for prosecution** in court.
- Enforcement is **sampling-based**, not exhaustive — officers work within a jurisdiction and prioritize inspections, they don't check every shop in the country. Our app's job is to help them prioritize smartly (e.g., surfacing shops with pending citizen complaints), not to replace this process.

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12. Final Team Decisions (Locked)

✅ **Track:** Student Innovation — SIH26197 (self-proposed, no fixed rubric) ✅ **Lead feature:** Food Safety (ingredient scanner, health scoring, complaint filing) ✅ **Legal Metrology:** Kept as a standalone consumer-facing check (2011 Rules as rule engine) — works independently, no officer data required ✅ **Officer section:** Kept by team choice for technical depth/impressiveness — scoped small (login + inspection logging + shop lookup only) ✅ **Reference PS:** SIH26034 used only as inspiration for the Legal Metrology module's scope — we are not bound by its rubric

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